



Exelon Generation.

Clinton Power Station
8401 Power Road
Clinton, IL 61727

U-604516

November 14, 2019

10 CFR 50.73
SRRS 5A.108

U.S. Nuclear Regulatory Commission
ATTN: Document Control Desk
Washington, D.C. 20555-0001

Clinton Power Station, Unit 1
Facility Operating License No. NPF-62
NRC Docket No. 50-461

Subject: Licensee Event Report 2019-003-00

Enclosed is Licensee Event Report (LER) 2019-003-00: Reactor Water Cleanup System Differential Flow Instrumentation Inoperable. This report is being submitted in accordance with the requirements of 10 CFR 50.73.

There are no regulatory commitments contained in this report.

Should you have any questions concerning this report, please contact Mr. Dale Shelton, Regulatory Assurance Manager, at (217) 937-2800.

Respectfully,

Thomas D. Chalmers
Site Vice President
Clinton Power Station

Attachment: Licensee Event Report 2019-003-00

cc:

Regional Administrator - Region III
NRC Senior Resident Inspector - Clinton Power Station
Office of Nuclear Facility Safety - Illinois Emergency Management Agency

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NRR



LICENSEE EVENT REPORT (LER)

(See Page 2 for required number of digits/characters for each block)

(See NUREG-1022, R.3 for instruction and guidance for completing this form
<http://www.nrc.gov/reading-rm/doc-collections/nuregs/staff/sr1022/r3/>)

Estimated burden per response to comply with this mandatory collection request 60 hours. Reported lessons learned are incorporated into the licensing process and fed back to industry. Send comments regarding burden estimate to the Information Services Branch (T-2 F43), U.S. Nuclear Regulatory Commission, Washington, DC 20555-0001, or by e-mail to InfoCollection.Resource@nrc.gov, and to the Desk Officer, Office of Information and Regulatory Affairs, NIOS-10202, (3150-0104), Office of Management and Budget, Washington, DC 20503. If a means used to impose an information collection does not display a currently valid OMB control number, the NRC may not conduct or sponsor, and a person is not required to respond to, the information collection.

1. Facility Name Clinton Power Station, Unit 1	2. Docket Number 05000461	3. Page 1 OF 3
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4. Title Reactor Water Cleanup System Differential Flow Instrumentation Inoperable

5. Event Date			6. LER Number			7. Report Date			8. Other Facilities Involved	
Month	Day	Year	Year	Sequential Number	Rev No	Month	Day	Year	Facility Name	Docket Number
09	16	2019	2019	003	00	11	14	2019	Facility Name	Docket Number 050000

9. Operating Mode 3	11. This Report is Submitted Pursuant to the Requirements of 10 CFR §: (Check all that apply)			
	☐ 20.2201(b)	☐ 20.2203(a)(3)(i)	☐ 50.73(a)(2)(ii)(A)	☐ 50.73(a)(2)(vii)(A)
	☐ 20.2201(d)	☐ 20.2203(a)(3)(ii)	☐ 50.73(a)(2)(ii)(B)	☐ 50.73(a)(2)(vii)(B)
	☐ 20.2203(a)(1)	☐ 20.2203(a)(4)	☐ 50.73(a)(2)(iii)	☐ 50.73(a)(2)(ix)(A)
	☐ 20.2203(a)(2)(i)	☐ 50.36(c)(1)(i)(A)	☐ 50.73(a)(2)(iv)(A)	☐ 50.73(a)(2)(x)
10. Power Level 000	☐ 20.2203(a)(2)(ii)	☐ 50.36(c)(1)(ii)(A)	☐ 50.73(a)(2)(v)(A)	☐ 73.71(a)(4)
	☐ 20.2203(a)(2)(iii)	☐ 50.36(c)(2)	☐ 50.73(a)(2)(v)(B)	☐ 73.71(a)(5)
	☐ 20.2203(a)(2)(iv)	☐ 50.46(a)(3)(ii)	☐ 50.73(a)(2)(v)(C)	☐ 73.77(a)(1)
	☐ 20.2203(a)(2)(v)	☐ 50.73(a)(2)(i)(A)	☒ 50.73(a)(2)(v)(D)	☐ 73.77(a)(2)(i)
	☐ 20.2203(a)(2)(vi)	☐ 50.73(a)(2)(i)(B)	☐ 50.73(a)(2)(vii)	☐ 73.77(a)(2)(ii)
	☐ 50.73(a)(2)(i)(C)	☐ Other (Specify in Abstract below or in NRC Form 366A)		

12. Licensee Contact for this LER	
Licensee Contact Mr. Dale Shelton, Regulatory Assurance Manager	Telephone Number (Include Area Code) (217) 937-2800

13. Complete One Line for each Component Failure Described in this Report									
Cause	System	Component	Manufacturer	Reportable to ICES	Cause	System	Component	Manufacturer	Reportable to ICES

14. Supplemental Report Expected ☐ Yes (If yes, complete 15. Expected Submission Date) ☒ No	15. Expected Submission Date Month: Day: Year:
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Abstract (Limit to 1400 spaces, i.e., approximately 14 single-spaced typewritten lines)

On 09/16/19, at 0817 CDT, at approximately 0% power during shutdown for refueling outage C1R19, the Division 1 and Division 2 Reactor Water Cleanup (RT) system differential flow instrumentation was declared inoperable due to failing downscale caused by flashing in the sensing lines that occurred during reactor cooldown. The Division 1 and 2 RT differential flow instrumentation were declared inoperable in accordance with Technical Specification (TS) 3.3.6.1. The cause of this event was unit shutdown procedures were not optimized to increase subcooling in the RT system during reactor cooldown. The system alignment was changed in accordance with procedures to increase subcooling and indication was restored by 0852 CDT on 09/16/19. Procedures governing unit shutdown will be revised to increase subcooling in the RT system during reactor cooldown to reduce the likelihood of flashing in the sensing lines. This condition met the reportability requirements of 10 CFR 50.73(a)(2)(v)(D), any event or condition that could have prevented the fulfillment of the safety function of structures or systems that are needed to mitigate the consequences of an accident.

**LICENSEE EVENT REPORT (LER)
CONTINUATION SHEET**

(See NUREG-1022, R 3 for instruction and guidance for completing this form
<http://www.nrc.gov/reading-rm/doc-collections/nuregs/staff/sr1022/r3/>)

Estimated burden per response to comply with this mandatory collection request: 80 hours. Reported lessons learned are incorporated into the licensing process and fed back to industry. Send comments regarding burden estimate to the Information Services Branch (T-2 F43), U.S. Nuclear Regulatory Commission, Washington, DC 20555-0001, or by e-mail to Infocollects.Resource@nrc.gov, and to the Desk Officer, Office of Information and Regulatory Affairs, NEOF-10202, (3150-0104), Office of Management and Budget, Washington, DC 20503. If a means used to impose an information collection does not display a currently valid OMB control number, the NRC may not conduct or sponsor, and a person is not required to respond to, the information collection.

1. FACILITY NAME	2. DOCKET NUMBER	3. LER NUMBER		
Clinton Power Station, Unit 1	05000461	YEAR 2019	SEQUENTIAL NUMBER - 003	REV NO. - 00

NARRATIVE**PLANT AND SYSTEM IDENTIFICATION**

General Electric -- Boiling Water Reactor, 3473 Megawatts Thermal Rated Core Power
Energy Industry Identification System (EIS) codes are identified in text as [XX].

EVENT IDENTIFICATION

Reactor Water Cleanup System Differential Flow Instrumentation Inoperable

A. Plant Operating Conditions Before the Event

Unit: 1	Event Date: September 16, 2019	Event Time: 0817
Mode: 3	Mode Name: Hot Shutdown	Reactor Power: 000

B. Description of Event

On September 16, 2019, at 0817 CDT, at approximately 0% power during shutdown for refueling outage C1R19, the Division 1 and Division 2 Reactor Water Cleanup (RT) [CE] system differential flow instrumentation [FI] were declared inoperable due to failing downscale caused by flashing in the sensing lines that occurred during reactor cooldown. The Division 1 and Division 2 RT differential flow instrumentation were declared inoperable in accordance with Technical Specification (TS) 3.3.6.1, Conditions D and E, which require restoring at least one division of instruments to Operable status within one hour. In response to this condition, the system alignment was changed in accordance with procedures to increase subcooling and restore indication. Division 1 and 2 RT differential flow instrumentation were declared Operable at 0852 CDT on September 16, 2019.

C. Cause of the Event

The plant was being shut down in accordance with the Unit Shutdown procedure (CPS 3006.01) when the Division 1 and Division 2 RT system differential flow instrumentation failed downscale. After the RT system differential flow instrumentation was declared inoperable, an Operator identified that the RT system operating procedure (CPS 3303.01) provides instructions to minimize flashing as reactor pressure is lowered during reactor cooldown. This procedure was used to restore the RT system to Operable status. Because the Unit Shutdown procedure does not direct the Operator to refer to guidance provided in the RT system operating procedure (CPS 3303.01) to minimize flashing as reactor pressure is lowered during plant cooldown, the cause of this event was determined to be

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CONTINUATION SHEET**

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Estimated burden per response to comply with this mandatory collection request 80 hours. Reported lessons learned are incorporated into the licensing process and fed back to industry. Send comments regarding burden estimate to the Information Services Branch (T-2 F43), U.S. Nuclear Regulatory Commission, Washington, DC 20555-0001, or by e-mail to InfoCollects.Resource@nrc.gov, and to the Desk Officer, Office of Information and Regulatory Affairs, NEOF-10202, (3150-0104), Office of Management and Budget, Washington, DC 20503. If a means used to impose an information collection does not display a currently valid OMB control number, the NRC may not conduct or sponsor, and a person is not required to respond to, the information collection.

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Clinton Power Station, Unit 1	05000461	YEAR 2019	SEQUENTIAL NUMBER - 003	REV NO. - 00

NARRATIVE

unit shutdown procedures were not optimized to increase subcooling in the RT system during cooldown, which resulted in flashing in the sensing lines.

D. Safety Consequences

The condition described in this LER rendered the Leakage Detection System incapable of performing its isolation function, which met the reportability requirements of 10 CFR 50.73(a)(2)(v)(D), any event or condition that could have prevented the fulfillment of the safety function of structures or systems that are needed to mitigate the consequences of an accident. There were no safety consequences associated with the event described in this report as the unit was in Mode 3 at the time of the event and the condition existed for less than thirty minutes, which is within the TS 3.3.6.1 Required Action time to restore at least one division of instruments to Operable status within one hour. In addition, manual isolation capability and automatic isolation capability in response to low reactor water level remained available.

E. Corrective Actions

- (1) The system alignment was changed in accordance with procedures to increase subcooling and restore indication. Division 1 and 2 RT differential flow instrumentation were declared Operable at 0852 CDT on September 16, 2019.
- (2) Procedures governing unit shutdown will be revised to better utilize existing guidance to increase subcooling in the RT system during reactor cooldown, which will reduce the likelihood of flashing in the sensing lines.

F. Previous Similar Occurrences

LER 2015-001-00 reported a loss of Division 1 and 2 RT differential flow instrumentation due to leak-by resulting in voids in the RT system that caused differential flow instrumentation to provide false readings. The actions taken for LER 2015-001-00 would not have prevented the condition described in this LER.

G. Component Failure Data

Not applicable.